

The active ingredient β -sitosterol in the anti-inflammatory agents alleviates perianal inflammation in rats by inhibiting the expression of Srebf2, activating the PPAR signaling pathway, and altering the composition of gut microbiota

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ABSTRACT

Background: Anti-inflammatory herbal formulations are common in traditional Chinese medicine for clearing heat and detoxifying; however, the specific active components and their mechanisms remain unclear.

Objective: This study investigates the role of Sitosterol in alleviating perianal inflammation and its underlying mechanisms.

Methods: Sitosterol was identified as a key active ingredient through the TCMSP database. Its structure was analyzed using PubChem, target genes were explored with STITCH, and KEGG pathways related to Srebf2 were revealed by STRING. An animal model of perianal inflammation was induced with 75 % acetic acid and treated with Sitosterol, water, normal saline, or antibiotics. The effects on gut microbiota were assessed using 16S rRNA sequencing, and inflammation was evaluated through HE stains, IHC, and TUNEL assays. In vitro, LPS-treated Caco-2 cells were used to measure proliferation, apoptosis, and cytokine levels, with PPAR pathway involvement examined using GW6471.

Results: Sitosterol emerged as the primary active ingredient targeting Srebf2, with KEGG analysis highlighting the PPAR signaling pathway. In rats, Sitosterol reduced weight loss, inflammatory cell infiltration, edema, and vasodilation in perianal tissue. Additionally, it decreased PCNA levels, increased apoptosis, and elevated serum levels of IL-1 β , IL-6, and TNF- α , particularly at high doses compared to antibiotics. Sitosterol also restored gut microbiota. Srebf2 knockdown improved tissue conditions and modulated cytokine levels, effects that were countered by GW6471. In LPS-treated Caco-2 cells, Sitosterol reversed reductions in cell viability and proliferation and modulated the expression of proteins and cytokines.

Conclusion: Sitosterol restores gut microbiota composition and further alleviates perianal inflammation in rats by inhibiting Srebf2 expression and activating the PPAR signaling pathway.

1. Introduction

Perianal inflammatory diseases include perianal abscesses and anal fistulas, which result from infections in the anal glands located between the small sphincters. These conditions involve complex

pathophysiologic mechanisms influenced by both local and systemic factors [1]. Perianal Crohn's disease (CD) is a specific type of perianal inflammatory disease that presents as abscesses and fistulas in localized areas of the perianal region [2]. Additionally, disturbances in the intestinal flora may contribute as a systemic factor in the development of

Abbreviation: Abbreviation, Full name; PPARs, peroxisome proliferator-activated receptors; Srebf2, Sterol regulatory element-binding protein 2; PPI, protein-protein interaction; HE, hematoxylin and eosin; LPS, Lipopolysaccharide; IHC, immunohistochemistry; PCoA, principal coordinate analysis.

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perianal inflammatory disease [3]. Antibiotics such as metronidazole are commonly used to control abscesses in the perianal area by reducing inflammation [4]. However, this approach can lead to antibiotic resistance, and symptoms often recur once the drug is discontinued [5]. This limitation hampers the ability to address the disease's root cause. Consequently, current research is focused on finding a novel therapeutic strategy that effectively alleviates inflammation while minimizing side effects.

In recent years, natural compounds of plant origin have become a hotspot in anti-inflammatory drug research due to their wide range of biological activities and low toxicity [6]. Sitosterol, also known as β -sitosterol, is a natural phytosterol that is most abundant in various plants, including fruits, vegetables, nuts [7,8].

Currently, numerous studies have investigated the anti-inflammatory effects of Sitosterol [9]. In 2023, several studies demonstrated that Sitosterol ameliorates inflammatory responses by inhibiting the NF- κ B signaling pathway [10–12]. Additionally, a study by Liyuan Ma et al. in 2023 found that Sitosterol could prevent allergic reactions by restoring the composition of gut microbiota in mice [13]. Consequently, it is speculated that Sitosterol may play a role in the treatment of perianal inflammatory diseases; however, its specific mechanism of action still requires further exploration.

Sterol Regulatory Element Binding Transcription Factor 2 (Srebf2), also known as Sterol Regulatory Element Binding Protein 2 (Srebp2), is a transcription factor essential for cholesterol homeostasis and lipid metabolism [14]. However, the maturation of Srebp2 activates NLRP3 inflammatory vesicles, which are involved in the development of multiple inflammatory diseases [15,16]. Research has shown that peroxisome proliferator-activated receptors (PPARs) can alleviate various inflammatory conditions by directing the differentiation of different immune cell types. Additionally, all three isoforms of PPAR (PPAR α , PPAR β/δ , and PPAR γ) have demonstrated anti-inflammatory properties [17,18]. Regarding the role of Sitosterol in Srebp2 and PPAR, Huan Qin et al. (2024) found that Sitosterol ameliorated lung inflammation caused by respiratory syncytial virus (RSV) infection in mice by inhibiting the cleavage maturation of Srebp2 [19]. In a separate study, Selvaraj Jayaraman et al. (2021) discovered that Sitosterol reduced SREBP-1c levels and enhanced PPAR- γ gene expression in the adipocytes of diabetic rats, helping to mitigate inflammation in these animals [20]. Although the roles of Sitosterol in both Srebp2 and PPAR have been addressed separately, the regulatory relationship between Srebp2 and PPAR remains unclear. Furthermore, literature indicates that cholesterol depletion activates Srebp2 [21], which in turn promotes PPAR expression through the early growth response-1 (EGR-1) pathway, further alleviating the inflammatory response. This finding contradicts the prevailing theory that PPAR has an anti-inflammatory effect [17,18]. Therefore, this paper will explore the role of Sitosterol in the treatment of perianal inflammation through Srebp2 and PPAR.

The intestinal flora consists of various microorganisms that reside in the digestive tract and play a vital role in maintaining host health [22]. Changes in gut flora, known as intestinal dysbiosis, are associated with a range of inflammatory conditions, including those affecting the anal region [23]. The gut microbiota influences systemic inflammation by producing microbial metabolites, modulating immune responses, and interacting with the gut barrier [24]. In addition, earlier studies have shown that an imbalance in the gut microbiota affects tryptophan metabolism and increases Srebp2 levels, potentially predisposing individuals to hepatocellular carcinoma [25]. Therefore, interventions that restore a healthy gut microbiota balance may offer therapeutic benefits in controlling inflammation.

Based on the research background outlined above, we investigated how Sitosterol regulates the expression of Srebf2 and activates the PPAR signaling pathway. Our goal was to elucidate the effects of Sitosterol on the composition of the gut microbiota and its potential to alleviate perianal inflammation in rats. This study aims to provide insights into novel therapeutic approaches for controlling inflammation and

promoting gut health.

2. Material and methods

2.1. Regents

Specific antibodies targeting Srebf2 (ab289003), Bax (ab32503), Bcl-2 (ab182858), PCNA (ab92552), PPAR (ab314112), and β -actin (ab7817) were purchased from Abcam (Cambridge, UK). The TUNEL Apoptosis Assay Kit (C1091), ELISA kits for IL-1 β (PI303), IL-6 (PI328), and TNF- α (PT516), CCK-8 kit (C0038), EdU Cell Proliferation Kit (C0071L), and Annexin V-FITC Apoptosis Detection Kit (C1062M) were sourced from Beyotime (Beijing, China). Sitosterol (s9889) was acquired from Merck (Darmstadt, Germany), and GW6471 was obtained from MCE (USA).

2.2. Screening of active ingredients in anti-inflammatory compound

Extracts showcasing anti-inflammatory properties and their respective target molecules were obtained from the Traditional Chinese Medicine Systems Pharmacology Database and Analytical Resource (TCMSP, accessible at <http://tcmspw.com/tcmsp.php>). In the quest to pinpoint prospective bioactive compounds, selection criteria were applied, encompassing oral availability (OB) of at least 30 %, a drug likeness (DL) score exceeding 0.18, and a half-life (HL) of no less than 4 h. Following this, the bioactive constituents were pinpointed. An interaction map depicting protein-protein connections (PPI) was constructed to visualize the active compounds and underscore the pivotal elements among them.

2.3. Search for target genes of Sitosterol

This investigation employed the STITCH database (<http://stitch.embl.de/>) to pinpoint the interactive relationships between Sitosterol and its associated target molecules, constructing a comprehensive Sitosterol-target interaction map. The molecular structure of Sitosterol was sourced from the PubChem repository (<https://pubchem.ncbi.nlm.nih.gov/>). Following this, data regarding the target genes of Sitosterol were utilized to generate an illustrative data table, which underwent further analysis with STRING to gain deeper insights. The keyword 'Sitosterol' was used to conduct a search in the STITCH database, uncovering the various targets linked to Sitosterol.

2.4. KEGG pathway and PPI network construction of Sitosterol targets

This study employed the STRING database to carry out KEGG pathway analysis. Following that, the STRING platform in conjunction with Cytoscape application was utilized to construct a network diagram showcasing the protein-protein interactions (PPIs) among the genes targeted by Sitosterol.

2.5. Cell culture and treatment

The Caco-2 cell line was maintained in a controlled environment at 37 °C with 5 % carbon dioxide in an incubation chamber. The cells proliferated in Dulbecco's Modified Eagle Medium (DMEM) from Gibco, Grand Island, United States, supplemented with 10 % fetal bovine serum (FBS), also provided by Gibco. Additionally, the culture medium was enriched with 100 units per milliliter of penicillin and 100 micrograms per milliliter of streptomycin, obtained from Beyotime (C0222), based in Beijing, China. Caco-2 cells were treated with solvent control, Lipopolysaccharide (LPS), LPS + 10 Mol/L-Sitosterol, LPS + 10 Mol/L-Sitosterol+sh-NC, or LPS + 10 Mol/L-Sitosterol+sh-Srebf2. In another in vitro experiment, Caco-2 cells were treated LPS + sh-NC, LPS + sh-Srebf2 or LPS + sh-Srebf2 + GW6471.

2.6. Transfection

To reduce the expression of Srebf2, Caco-2 cells were transfected with 10 nanomoles of Srebf2-specific short hairpin RNA (shRNA) along with a control shRNA, both obtained from GenePharma in Shanghai, China, using Lipofectamine 2000 reagent. After 24 h, the transfected cells were prepared for experimental applications. The specific sequence used for sh-Srebf2 in this study was: 5'-GATCCGCCATGACC-CATCTGTTGAGCTTCAAGAGAGCTCAACA-GATGGGTCATGGCTTTTGGAAA-3'.

2.7. Establishment of stable knock-down cell lines

We utilized the GV355 lentiviral shRNA construct (hU8-MCS-Ubiquitin-fluorescent Luciferase-IRES-neomycin) sourced from GeneChem, based in Shanghai, China, to infect both Caco-2 cell cultures and those exposed to LPS. Subsequently, we purified and established cell lines that exhibited stable depletion of Srebf2.

2.8. Animals and treatments

Following a 7-day adjustment period, 54 adult SD rats (8 weeks, 180 ± 20 g) were maintained in conditions with a 12-h light/dark cycle at 26 ± 2 °C and 55 ± 10 % relative humidity. The rats were kept in a 1:1 male-to-female ratio, with unrestricted access to food and water. To create the perianal inflammation animal model, the perianal area of the SD rats (body weight 180 ± 20 g) was cleaned with iodine tincture (Betadine Tincture, BET-100, USA). Following this, 0.05 mL of 75 % ice-cold acetic acid (Sigma-Aldrich, A6283, USA) was administered via subcutaneous injection around the rat's anus [26,27]. After 24 h, the presence of ulcers around the anus was checked, with signs of redness, swelling, and inflammatory exudate, confirming the successful establishment of the model.

Using the rat perianal inflammation model, we carried out animal study I. In this investigation, the rats were randomly allocated into six groups ($n = 6/\text{group}$): negative control (rats given water), model group (rats administered normal saline), treatment 1 (model rats receiving a low dose of Sitosterol (1 mg/kg)), treatment 2 (model rats given a medium dose of Sitosterol (5 mg/kg)), treatment 3 (model rats treated with a high dose of Sitosterol (100 mg/kg)), and positive control (rats given antibiotics). From the day after modeling, each group was given the respective experimental drugs by gavage three times a day, with a 4-h interval between doses, for a total of 16 consecutive days. During the treatment phase, the rats' body weight was continuously monitored.

In animal study II, a suspension of roughly 2 million Caco-2 cells (derived from the sh-Srebf2 cell line) was prepared in a volume of 100 µl of phosphate-buffered saline. This suspension was then injected into the perianal region of the rodent subjects. Following this procedure, the rodents were divided at random into three distinct cohorts, each consisting of six animals: a control cohort, a cohort receiving the model treatment plus a sh-NC intervention, and a final cohort receiving the model treatment alongside the sh-Srebf2 intervention.

Ultimately, the rats were sacrificed with 4 % isoflurane (Abbott, USA). Following euthanasia, perianal tissues were gathered for histopathological and molecular biology assessments. Additionally, fecal samples were collected for 16 s RNA sequencing.

2.9. Microbial genomic DNA extraction and high throughput sequencing

Fecal microbial community genomic DNA was isolated utilizing the E.Z.N.A.® Soil DNA Extraction Kit (produced by Omega Bio-tek in the United States). After isolation, the DNA's integrity was evaluated through 1 % agarose gel electrophoresis, while its concentration was determined using a NanoDrop 2000 UV-Vis Spectrophotometer (manufactured by Thermo Scientific, also from the United States). Amplification of the V3-V4 hypervariable regions of the 16S rRNA gene was

carried out using the primers 338F (5'-ACTCGTACGGGAGTCAGCAG-3') and 806R (5'-GGTCTACHVAGGTWTCTAAT-3'), via an ABI GeneAmp®9700 PCR machine (provided by ABI, United States). The PCR protocol comprised an initial denaturation at 95 °C for 3 min, followed by 27 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, and extension at 72 °C for 45 s, culminating with a final extension at 72 °C for 10 min. The PCR master mix contained TransStart FastPfu buffer, deoxynucleotide triphosphates (dNTPs), the primers, TransStart FastPfu DNA Polymerase, template DNA, and distilled water, to achieve a total volume of 20 µl. Each PCR reaction was performed in triplicate. The resultant PCR products were then purified with the AxyPrep DNA Gel Extraction Kit (from Axygen Biosciences, USA) and their concentrations measured using a Quantus™ Fluorometer (produced by Promega, USA). After purification, the amplicons were combined in equal molar ratios and subjected to sequencing on the Illumina MiSeq PE300/NovaSeq PE250 systems (Illumina, USA) by Majorbio Bio-Pharm Technology Co. Ltd., based in Shanghai, China.

2.10. Histological examination

Rectal periosteal samples were procured and embedded in a beeswax matrix. The samples were then meticulously cut into 5-µm thick sections. These sections underwent a dewaxing process, followed by rehydration and staining with the hematoxylin-eosin (HE) technique. The detection of apoptotic cells was performed using the TUNEL method, following the guidelines provided by the TUNEL Apoptosis Assay Kit (Product Code C1091) manufactured by Beyotime, based in Beijing, China. The sections stained with HE and TUNEL were imaged using the Panoramic MIDI whole-slide scanner, produced by 3DHISTECH Ltd., situated in Budapest, Hungary, at 20× zoom. Subsequent analysis was conducted with the Panoramic Viewer software, also developed by 3DHISTECH in Budapest, Hungary.

2.11. ELISA

The levels of interleukin-1 beta (IL-1β), interleukin-6 (IL-6), and tumor necrosis factor-alpha (TNF-α) in the rat bloodstream and the supernatant of cell cultures were quantified using enzyme-linked immunosorbent assay (ELISA) reagent sets manufactured by Beyotime, based in Beijing, China, following the manufacturer's instructions. The optical density was measured with a microplate spectrophotometer (Model 3550 by Bio-Rad) at the 450 nm spectral range for analysis purposes.

2.12. Cell proliferation

The assessment of cellular multiplication was conducted using the CCK-8 analysis system (product code C0038, supplied by Beyotime, Beijing, China). The cells were seeded at a density of 5000 per well and cultured in a 96-well dish containing 100 µl of complete DMEM medium. After this, 10 µl of the CCK-8 solution were added to each well. The absorbance at the 450 nm wavelength was measured using a Microplate Reader (manufactured by Bio-Rad, Hercules, California, USA) after a 24-h incubation following treatment.

2.13. EdU assay

The assessment of cell multiplication was conducted using the EdU detection kit (product code C0071L, supplied by Beyotime, Beijing, China). The cells were seeded at a density of 1 million per well and treated, followed by a 2-h incubation with the EdU reagent. After treatment, the cells were fixed and permeabilized before being subjected to the click reaction buffer and stained with DAPI. Fluorescent images were captured using a fluorescence microscope (model Olympus BX63, manufactured in Japan), and the analysis for determining the rate of cellular growth was performed with ImageJ software (version 1.8.0, developed by the National Institutes of Health, Bethesda, Maryland,

USA).

2.14. Cell apoptosis analysis

For the examination of cellular apoptosis, the cells were seeded into 6-well culture dishes at a density of approximately 1 million cells per well. Apoptosis was assessed using the Annexin V-FITC kit (product code C1062M, produced by Beyotime, China), following the manufacturer's protocol closely.

3. qRT-PCR

The extraction of RNA was performed using cellular or tissue samples with Trizol (manufactured by Beyotime, Beijing, China). Subsequently, complementary DNA was synthesized using a specialized cDNA synthesis kit, also provided by Beyotime. The PCR amplification was conducted on an ABI 7900 high-throughput real-time PCR system (produced by ABI, United States). Below are the sequences of the primers used: Srebf2 (rat): 5'-TGCAGGATCAGAGGAGGAG-3' and 5'-GTTGTTGCCG TGATGGTGAC-3'; Srebf2 (human): 5'-GCTGGTGCTTCTGTTTCTG-3' and 5'-GAGACAGCCTCCTGTTGCTG-3'; Bax (rat): 5'-AGCCTGAGAGG ATGATTGTTG-3' and 5'-TGTGGAGAGGAGCAGGAGTG-3'; Bax (human): 5'-GAGGATGATTGCCGCCGAG-3' and 5'-GGCAGGAGGACC TGGGAGT-3'; Bcl-2 (rat): 5'-TACGCCAGCCTGACGCCG-3' and 5'-CAG-CAGGCGATGACGAGG-3'; Bcl-2 (human): 5'-ATGCCCTTGAGTTCGCCG-3' and 5'-TCTTTGAGTTCTCCCGG-3'; PCNA (rat): 5'-ACCTGAGCCT-GATGCTGTC-3' and 5'-AGCAGCTCAGCCTCCTGT-3'; PCNA (human): 5'-CAGCCAGACATCAGCAGCAC-3' and 5'-TCTGAGCAGCAGGACCTT GA-3'; PPAR (rat): 5'-CGGCTGTTTACAGCAGACAG-3' and 5'-GCTG GTGCTGCTGCAATCTC-3'; PPAR (human): 5'-CCCTCTCTCCTCCCTT CTTG-3' and 5'-GCTGCTGGTCAGAGGATGAC-3'; β -actin (rat): 5'-CAC-CATTGGCAATGACGGTTC-3' and 5'-AGGTCTTTGCCGATGTCCACGT-3'; β -actin (human): 5'-GTGGGGCGCCCCAGGCACCA-3' and 5'-CTCCTTAATGTACGCACGATTTC-3'. The protocol commenced with a preliminary denaturation phase at 95 °C for 10 min. This was followed by 40 cycles, each consisting of a denaturation step at 95 °C for 10 s, an annealing step at either 52.8 °C or 56.2 °C for 15 s, and a final extension phase at 72 °C for 20 s. A thermal gradient was applied, increasing from 72 °C to 95 °C by 1 °C at each interval. Automatic gain calibration was performed before the first operation. For mRNA quantification, the expression of β -actin was used as a standard internal control, employing the 2^{- $\Delta\Delta$ Ct} methodology.

3.1. Western blot

Samples were disrupted using RIPA lysis buffer (Product Code P0013B, Beyotime Institute, Beijing, P.R.C.). Protein concentrations were assessed with a ready-to-use assay kit (Product Code P0006, Beyotime Institute, Beijing, P.R.C.). After extraction, proteins were electrophoretically transferred to PVDF sheets, and their binding sites were blocked with a 5 % non-fat milk solution. A microbial inhibition compound was then applied to the PVDF sheets (Product Code IPVH00010, Millipore Corporation, United States) and allowed to react at 4-degree Celsius overnight. Post this, the membranes were subjected to probing with human or rodent polyvalent antibodies targeting Srebf2 (dilution 1:1000, Catalogue Number ab289003, Abcam, United Kingdom), Bax (dilution 1:1000, Catalogue Number ab32503, Abcam, United Kingdom), Bcl-2 (dilution 1:2000, Catalogue Number ab182858, Abcam, United Kingdom), PCNA (dilution 1:2000, Catalogue Number ab92552, Abcam, United Kingdom), PPAR (dilution 1:1000, Catalogue Number ab314112, Abcam, United Kingdom), and β -actin (dilution 1:1000, Catalogue Number ab7817, Abcam, United Kingdom). This was succeeded by a treatment with horseradish peroxidase-conjugated Goat Anti-Human or Goat Anti-Rat secondary antibodies (Product Codes A0201 or A0192, dilution 1:1000, Beyotime Institute, Beijing, P.R.C.). The protein bands were made visible through the application of an

enhanced chemiluminescence reagent. β -actin was used as a normalization control, and the relative protein quantities were analyzed using densitometric measurements with ImageJ software.

3.2. Statistical analysis

Statistical computations were conducted using SPSS software, version 22. Continuous data were analyzed by calculating the mean and determining the standard deviations. To assess differences between two distinct groups, an independent samples *t*-test was performed. For comparing variances across three or more groups, a one-factor ANOVA was deployed. Any *p*-value less than 0.05 was considered to indicate statistical significance.

4. Results

4.1. Sitosterol is the key active ingredient in the anti-inflammatory compound

In our study, the anti-inflammatory compound is a traditional Chinese medicine oral preparation that includes 9 g of honeysuckle, 9 g of *Forsythia suspensa*, 9 g of peony bark, 9 g of red peony, 6 g of soapberry thorn, 30 g of purple-flowered dandelion, 30 g of dandelion, 12 g of large green leaves, 3 g of dried tangerine peel, and 5 g of raw licorice. To identify the active ingredients of this anti-inflammatory compound, we utilized TCMS to search for these components. Fig. 1A illustrates the PPI network of the traditional Chinese medicine components in the anti-inflammatory compound along with their active ingredients. The radar chart in Fig. 1B displays the number of active ingredients found in various traditional Chinese medicines within the anti-inflammatory compound, highlighting that licorice contains the highest number of active ingredients.

We further filtered the data with a degree ≥ 2 , identifying 13 potential active ingredients. Sitosterol was found to have the highest degree, indicating that it is the most crucial active component in the anti-inflammatory compound (Fig. 1C). We queried the structure of Sitosterol using the PubChem database (Fig. 1D). The STITCH database was then employed to analyze the targets of Sitosterol, resulting in the identification of 10 related genes (Fig. 1E). Among these, Srebf2 had the highest combined score (Fig. 1F), suggesting that Srebf2 is a key target gene of Sitosterol.

After conducting KEGG pathway enrichment studies, we observed a significant accumulation of Srebf2 and its associated genes within the PPAR signaling pathway (Fig. 1G). This suggests that the PPAR signaling pathway may serve as a crucial mechanism in the therapeutic action of Sitosterol. Essentially, our research has identified Sitosterol as the primary effective component in the anti-inflammatory mixture, which directly impacts Srebf2, a key player in the PPAR signaling cascade.

4.2. Sitosterol improves perianal inflammation and downregulates Srebf2 levels in rats

Sitosterol was used to treat perianal inflammation in a rat model, with antibiotics serving as the positive control. We found that body weight decreased in the perianal inflammation model ($P < 0.05$) but was restored by Sitosterol or antibiotics (Fig. 2A). HE and TUNEL staining revealed inflammatory cell infiltration, edema, and vasodilation (Fig. 2B), as well as increased cell apoptosis (Fig. 2C) in the perianal tissue of model rats, which were reversed by Sitosterol or antibiotics. Serum levels of IL-1 β , IL-6, and TNF- α were significantly elevated in model rats ($P < 0.001$) but decreased significantly with Sitosterol or antibiotics ($P < 0.001$, Fig. 2D). Furthermore, Srebf2, which is involved in the formation of inflammation, was found to have increased expression in model rats ($P < 0.001$) but was reduced by Sitosterol or antibiotics ($P < 0.001$, Fig. 2E, F), as demonstrated by real-time PCR and Western blot. In conclusion, these results indicate that rats with perianal

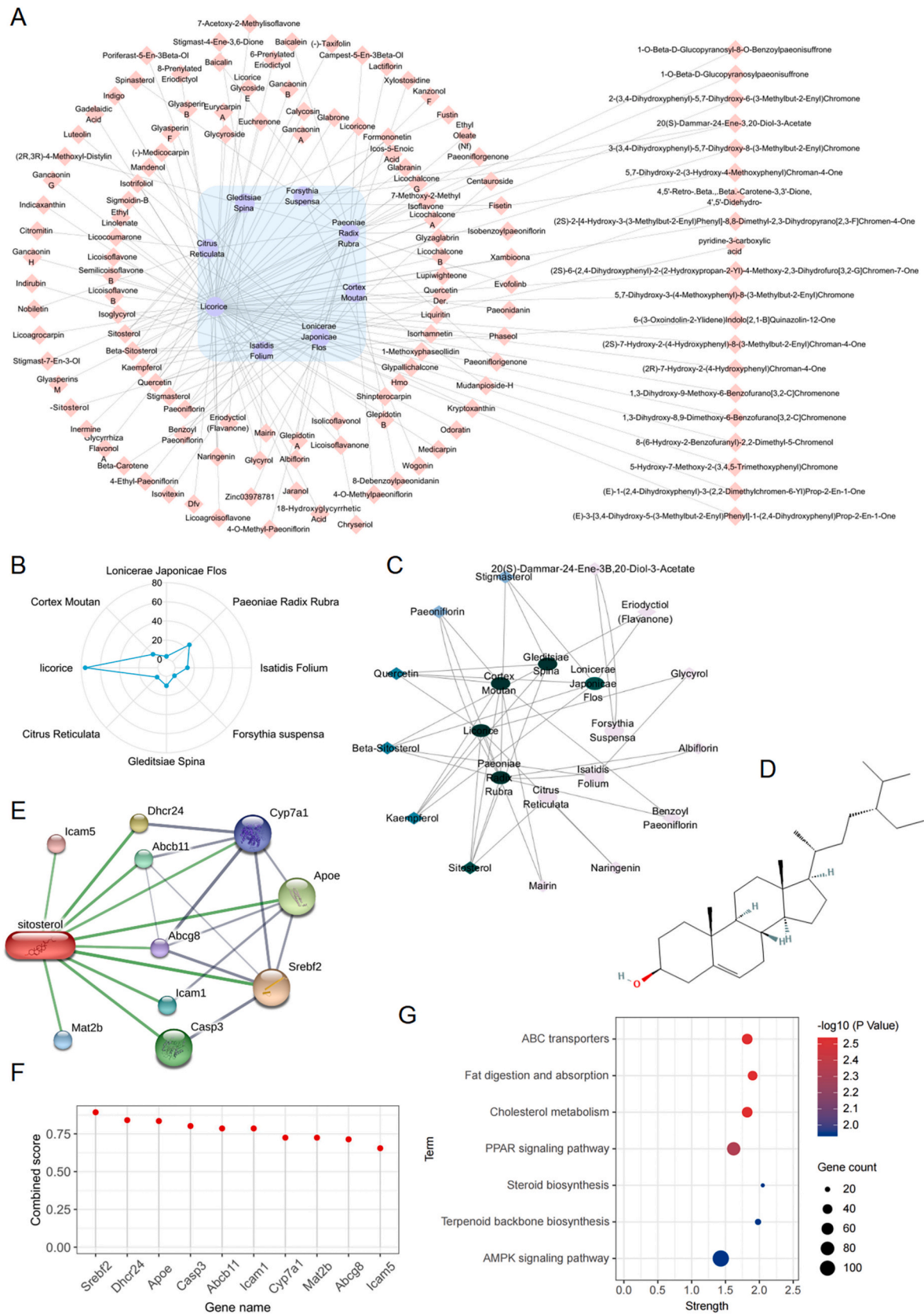


Fig. 1. Sitosterol is the key active ingredient in the anti-inflammatory compound. (A) Protein-protein interaction (PPI) network of traditional Chinese medicine components and their active ingredients. (B) Radar chart showing the number of active ingredients in each component, with licorice having the most. (C) Degree-based filtering reveals Sitosterol as the most crucial active ingredient. (D) Structure of Sitosterol from the PubChem database. (E) STITCH database analysis identifying 10 genes related to Sitosterol. (F) Srebf2 has the highest combined score as a key target gene. (G) KEGG pathway enrichment analysis shows Srebf2, and related genes are enriched in the PPAR signaling pathway.

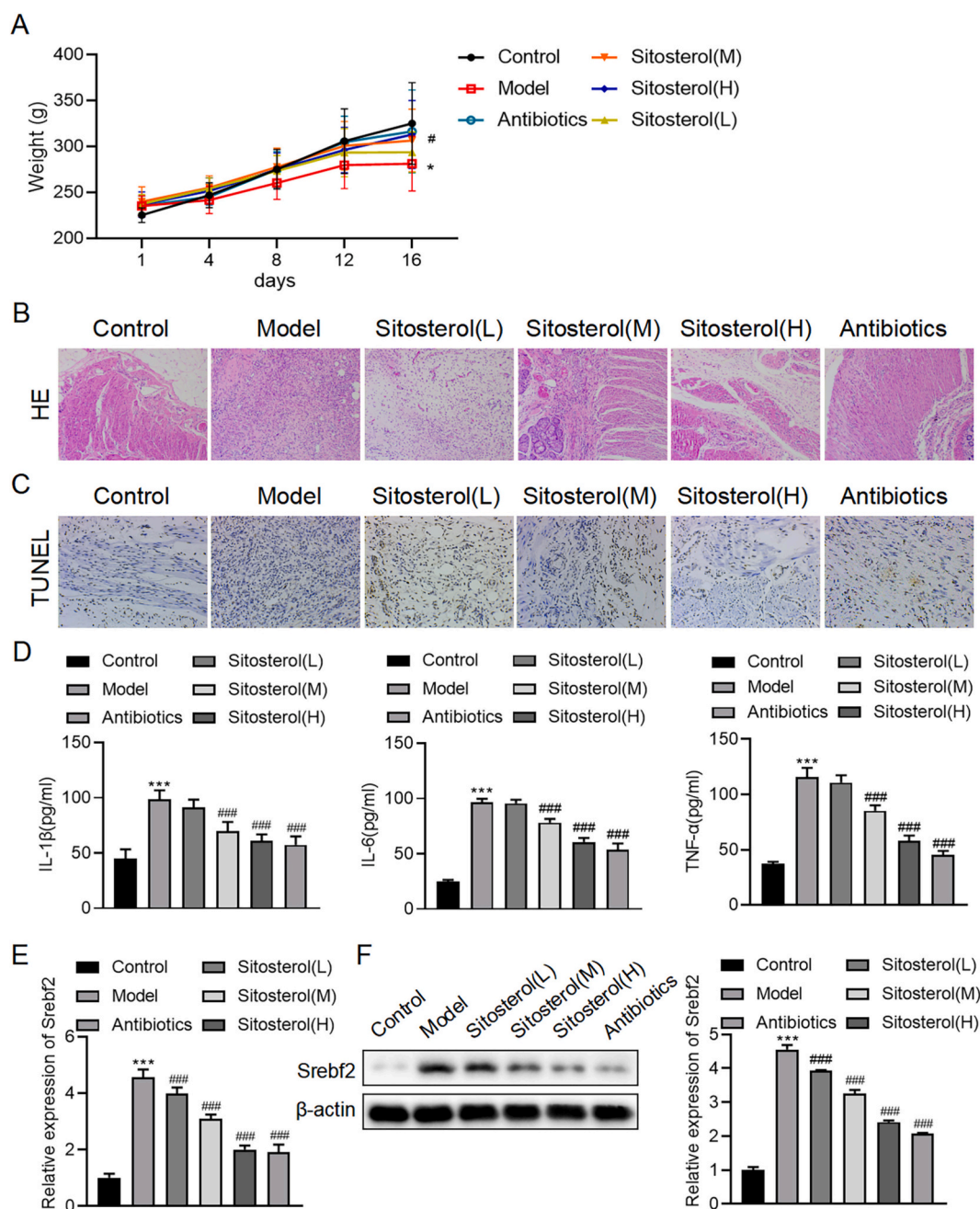


Fig. 2. Sitosterol improves perianal inflammation and downregulates Srebf2 levels in rats. (A) Body weight changes in perianal inflammation rats treated with Sitosterol or antibiotics, showing recovery with both treatments. (B) Pathological analyses revealing inflammatory cell infiltration, edema, and vasodilation, which are mitigated by Sitosterol or antibiotics by HE staining. (C) Increased cell apoptosis in model rats, reduced by Sitosterol or antibiotics by TUNEL. (D) Serum levels of IL-1 β , IL-6, and TNF- α significantly elevated in model rats but decreased with Sitosterol or antibiotics. (E-F) Real-time PCR and Western blot results showing increased Srebf2 expression in model rats, reduced by Sitosterol or antibiotics. Data are presented as mean $\pm$ standard deviation (SD). $n = 3$. *** $P < 0.001$ vs control group; ### $P < 0.001$ vs model group.

ulcers exhibit typical pathological features of an inflammatory response, characterized by an overall increase in inflammatory factors. Sitosterol effectively ameliorates inflammation in these rats.

4.3. Sitosterol promotes the recovery of gut microbiota in rats with perianal inflammation

To investigate the relationship between Sitosterol's role in alleviating inflammation in rats and the composition of the intestinal flora, we employed 16S rRNA sequencing to evaluate the influence of Sitosterol on the gut microbiota in rats with perianal inflammation. Principal

coordinates analysis (PcoA) clearly differentiates among the Control group, the Model group, treatment 1 (Sitosterol at 1 mg/kg), treatment 2 (Sitosterol at 5 mg/kg), treatment 3 (Sitosterol at 100 mg/kg), and the positive control group (Fig. 3A). The heatmap illustrates that samples from the six groups-negative control, model group, treatment 1 (Sitosterol at 1 mg/kg), treatment 2 (Sitosterol at 5 mg/kg), treatment 3 (Sitosterol at 100 mg/kg), and positive control are relatively like one another (Fig. 3B). PERMANOVA analysis revealed significant differences among the groups ($P < 0.001$, Fig. 3C). These findings suggest that Sitosterol may facilitate the restoration of gut microbiota in rats with perianal inflammation.

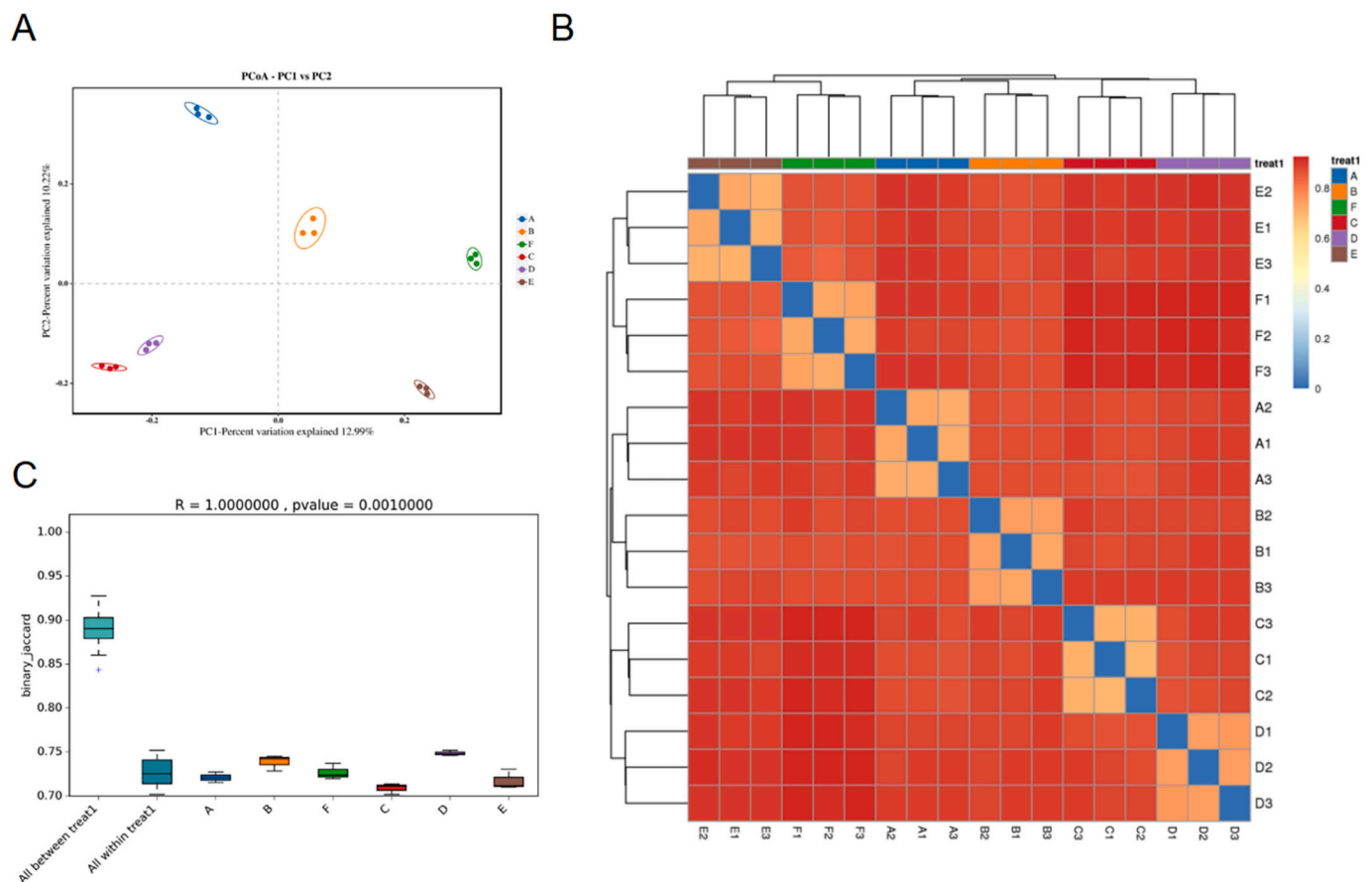


Fig. 3. Sitosterol promotes the recovery of gut microbiota in rats with perianal inflammation. (A) PCoA based on unweighted UniFrac distances showing significant differences in gut microbiota community structure among Control, Sitosterol, and antibiotics-treated groups. (B) Heatmap is based on distance algorithms (binary, bray, weighted, unweighted) to obtain the distance matrix between samples. (C) PERMANOVA analysis.

4.4. Knockdown of Srebf2 upregulates PPAR, restores gut flora composition, and alleviates perianal inflammation in rats

To further investigate the relationship between Srebf2 and inflammation in rats with perianal ulcers, we examined the role of Srebf2 in alleviating perianal inflammation by knocking down Srebf2 in a rat model of perianal inflammation. Realtime PCR and Western blot analyses showed that Srebf2 expression was significantly increased in the model rats (Fig. 4A, B, $P < 0.001$), while it decreased after transfection with sh-Srebf2 ($P < 0.001$), suggesting that Srebf2 is significantly knocked down in vivo. HE and TUNEL staining revealed inflammatory cell infiltration, edema, and vasodilation (Fig. 4C) as well as increased cell apoptosis (Fig. 4D) in the perianal tissue of the Model+sh-NC group, effects that were mitigated by sh-Srebf2. After knocking down Srebf2, we observed a significant downregulation of the pro-apoptotic protein Bax and an upregulation of the apoptosis inhibitor protein Bcl-2 ($P < 0.001$, Fig. 4E). Additionally, PPAR protein levels, which have a protective effect against inflammation, were also elevated (Fig. 4E). Additionally, serum levels of IL-1 β , IL-6, and TNF- α were reduced in model rats transfected with sh-Srebf2 ($P < 0.001$, Fig. 4F). These results indicate that the knockdown of Srebf2 in vivo in rats led to a significant improvement in inflammation in ulcerated rats.

Continuing to explore the relationship between Srebf2 and rat intestinal flora, we observed differences in the composition of the intestinal flora following the knockdown of Srebf2. The PcoA results indicated that the Model, Model+sh-NC, and Model+sh-Srebf2 groups could be clearly differentiated (Fig. 4G). The heat map demonstrates that samples from the Model, Model+sh-NC, and Model+sh-Srebf2 groups are closely clustered together (Fig. 4H). PERMANOVA analysis revealed significant differences among the groups ($P < 0.001$, Fig. 4I).

These results suggest that the knockdown of Srebf2 in vivo in rats restores the composition of intestinal flora and alleviates perianal inflammation.

4.5. Sitosterol promotes LPS-induced cell proliferation, inhibits apoptosis, and reduces the secretion of inflammatory factors in Caco-2 cells by inhibiting Srebf2

To investigate the relationship between Sitosterol and Srebf2 in reducing perianal inflammatory effects, we exposed LPS-treated Caco-2 cells (an in vitro model of perianal inflammation) to Sitosterol and transfected them with or without sh-Srebf2. First, we employed the CCK-8 assay to evaluate the impact of Sitosterol on the viability of these LPS-treated Caco-2 cells. The viability of LPS-treated Caco-2 cells was significantly reduced ($P < 0.001$), and elevated by Sitosterol treatment, particularly at the higher dose after 24 or 48 h ($P < 0.001$, Fig. 5A). Consequently, a high dose of Sitosterol was used in subsequent studies. After transfection with sh-Srebf2, Srebf2 expression was significantly reduced, as demonstrated by Realtime PCR and Western blot analyses ($P < 0.001$, Fig. 5B), indicating that our intracellular knockdown of Srebf2 was successful. Fig. 5C illustrates that Srebf2 levels were elevated in LPS-treated Caco-2 cells ($P < 0.001$), but Sitosterol reduced LPS-induced elevated Srebf2 levels, and sh-Srebf2 further reduced Srebf2 levels ($P < 0.001$). Cell proliferation, which was significantly diminished in LPS-treated Caco-2 cells ($P < 0.001$), was restored by Sitosterol, while sh-Srebf2 further restored cell proliferation, as shown by the EdU assay ($P < 0.001$, Fig. 5D). LPS-treated Caco-2 cells exhibited a notable increase in apoptosis, characterized by elevated Bax levels and reduced Bcl-2 levels ($P < 0.001$), but Sitosterol inhibited LPS-induced apoptosis, which was further inhibited by sh-Srebf2 ($P < 0.001$, Fig. 5E, F).

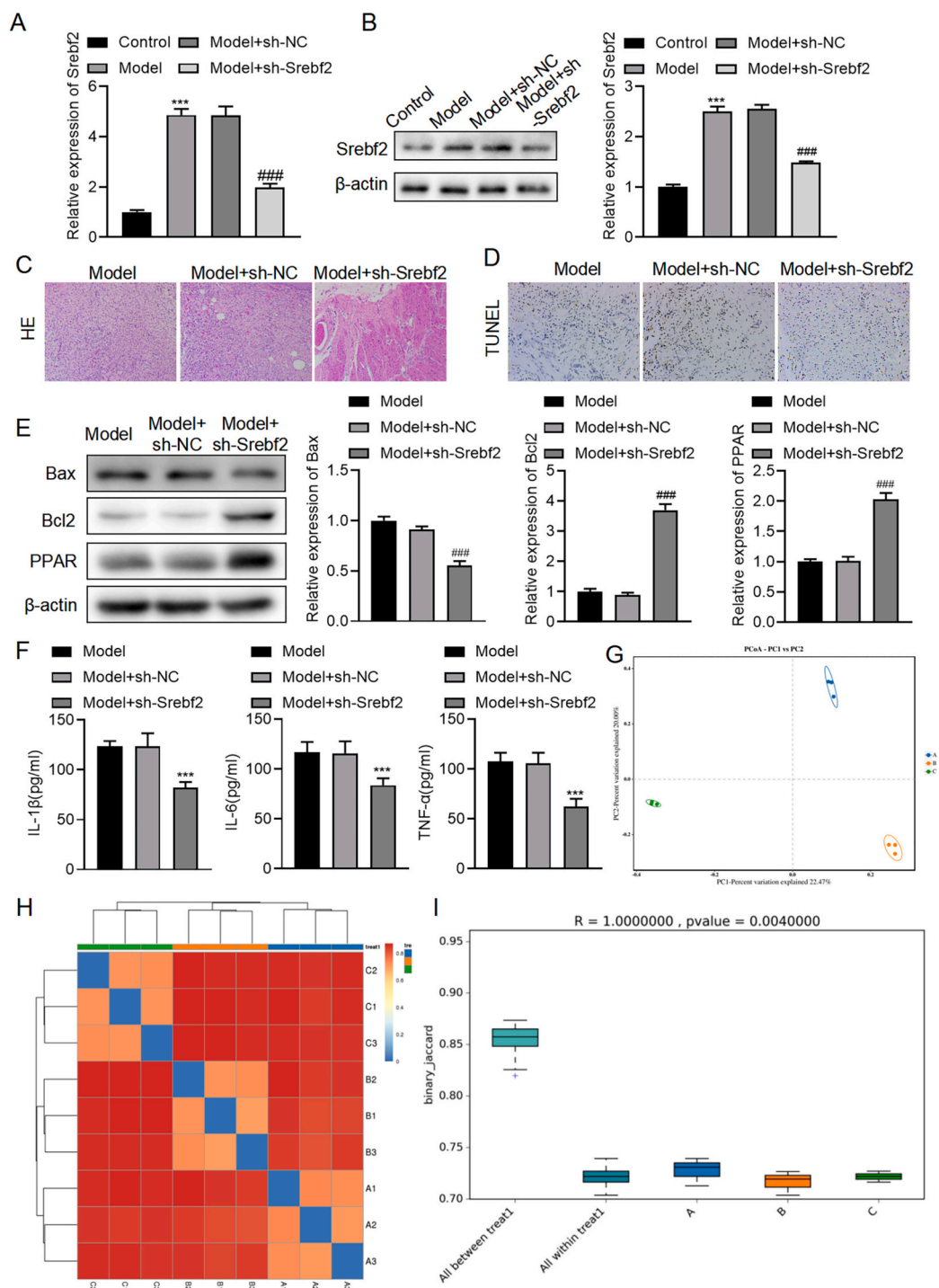


Fig. 4. Knockdown of Srebf2 upregulates PPAR, restores gut flora composition, and alleviates perianal inflammation in rats. (A-B) Real-time PCR and Western blot analyses showing increased Srebf2 expression in model rats, which was reduced following sh-Srebf2 transfection. (C) Pathological analyses in perianal tissue post sh-Srebf2 treatment by HE staining. (D) Reduced cell apoptosis in perianal tissue after sh-Srebf2 transfection by TUNEL. (E) Western blot results showing down-regulation of Bax and upregulation of Bcl-2, and PPAR following sh-Srebf2 treatment. (F) Reduced serum levels of IL-1 β , IL-6, and TNF- α in model rats transfected with sh-Srebf2. (G) PCoA based on unweighted UniFrac distances showing significant differences in gut microbiota community structure among Model, Model+sh-NC, and Model+sh-Srebf2 groups. (H) Heatmap is based on distance algorithms (binary, bray, weighted, unweighted) to obtain the distance matrix between samples. (I) PERMANOVA analysis. Data are presented as mean $\pm$ standard deviation (SD). $n = 3$. *** $P < 0.001$ vs Control group; ### $P < 0.001$ vs Model+sh-NC group.

Additionally, the levels of PCNA, a cell proliferation marker protein, and PPAR protein, which has an ameliorative effect on inflammation, were reduced in LPS-treated Caco-2 cells ($P < 0.001$), but Sitosterol restored these levels, sh-Srebf2 further restored the levels of both proteins ($P < 0.001$, Fig. 5F). The inflammatory markers IL-1 β , IL-6, and TNF- α were significantly elevated in LPS-treated Caco-2 cells ($P < 0.001$), but these

levels were markedly reduced by Sitosterol treatment, sh-Srebf2 further reduced the levels of inflammatory factor ($P < 0.001$, Fig. 5G). In conclusion, in LPS-treated Caco-2 cells, Sitosterol promotes cell proliferation, inhibits apoptosis, restores PPAR levels, and reduces inflammatory factor secretion by inhibiting Srebf2.

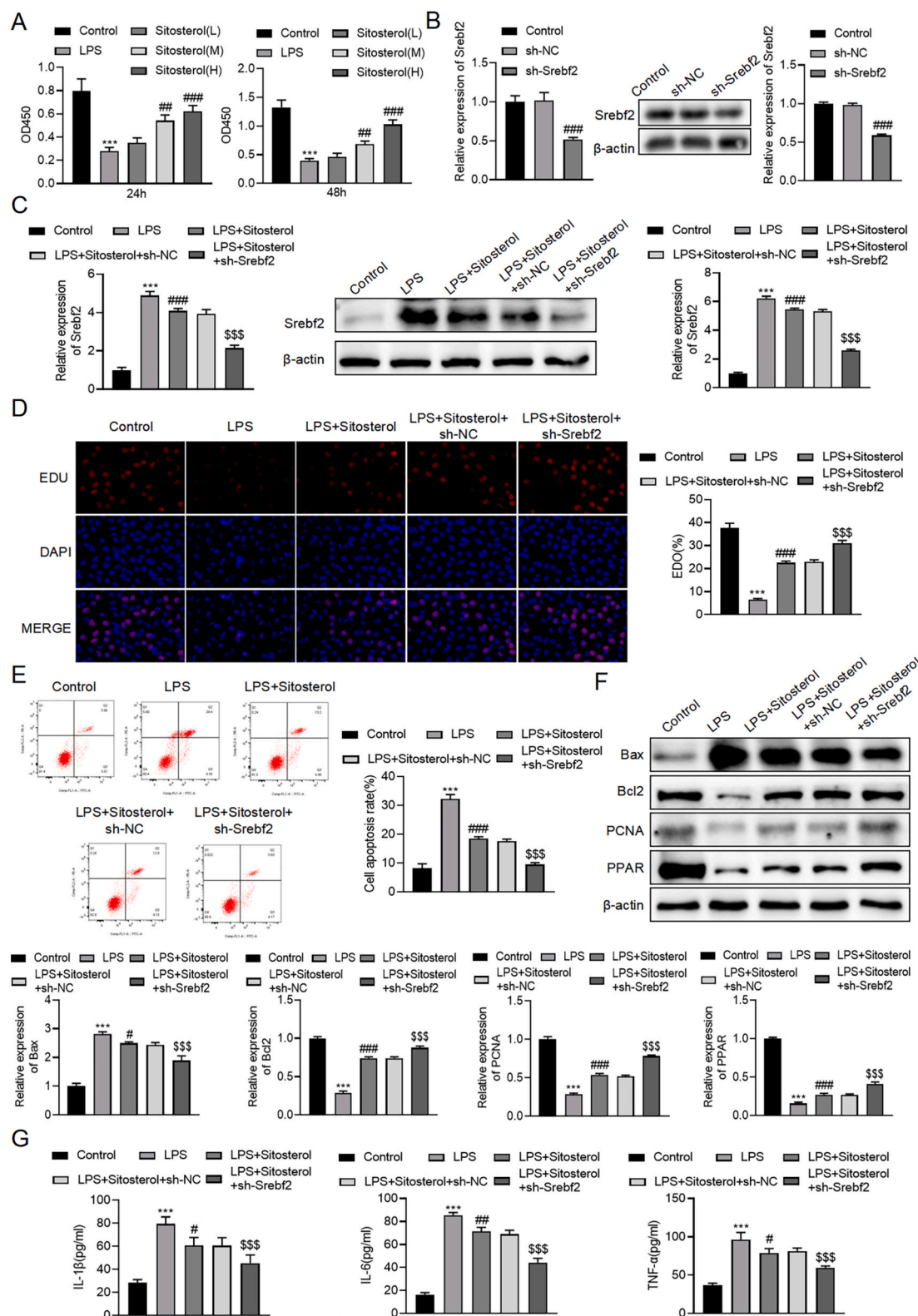


Fig. 5. Sitosterol promotes LPS-induced cell proliferation, inhibits apoptosis, and reduces the secretion of inflammatory factors in Caco-2 cells by inhibiting Srebf2. (A) CCK-8 assay shows that Sitosterol reverses the reduced cell proliferation in LPS-treated Caco-2 cells. (B) Realtime PCR and Western blot confirm significant reduction in Srebf2 expression following sh-Srebf2 transfection. (C) Srebf2 levels are elevated in LPS-treated cells but reduced by Sitosterol; sh-Srebf2 exacerbates this reversal. (D) EdU assay shows restored cell proliferation with Sitosterol, with worsened effects by sh-Srebf2. (E, F) Sitosterol reverses increased apoptosis and altered Bax/Bcl-2 ratios in LPS-treated cells; sh-Srebf2 worsens these effects. Sitosterol also restores PCNA and PPAR levels, which are reduced in LPS-treated cells. (G) Inflammatory factors IL-1 β , IL-6, and TNF- α are elevated in LPS-treated cells but significantly reduced by Sitosterol; sh-Srebf2 further exacerbates the decrease. Data are presented as mean $\pm$ standard deviation (SD). $n = 3$. *** $P < 0.001$ vs Control group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs LPS group or sh-NC; \$\$\$ $P < 0.001$ vs LPS + Sitosterol+sh-NC group.

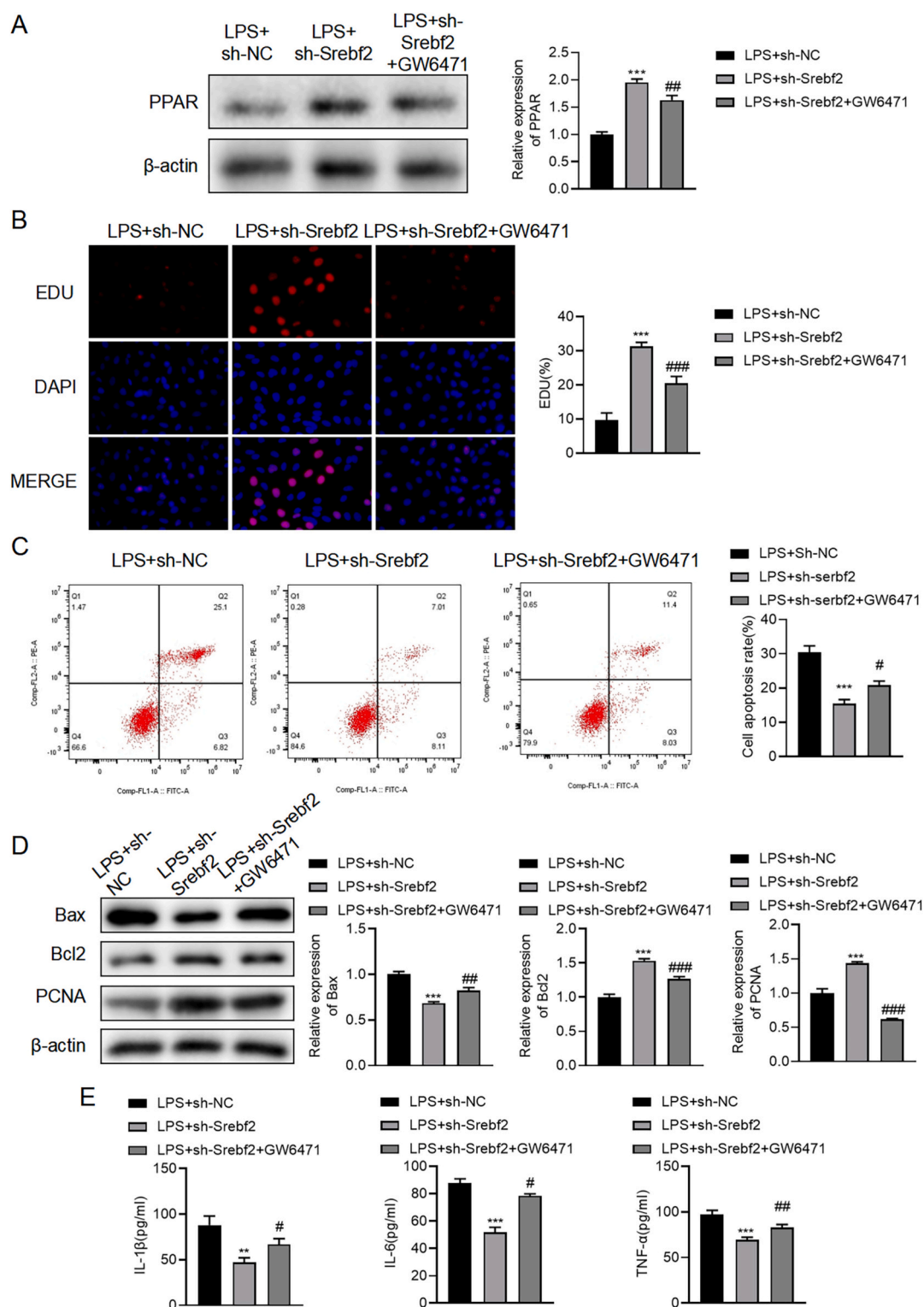


Fig. 6. Srebf2 inhibits cell proliferation, promotes apoptosis, and facilitates the secretion of inflammatory factors by inhibiting the PPAR signaling pathway. (A) PPAR levels are elevated in LPS-treated Caco-2 cells with sh-Srebf2 but are reversed by GW6471 treatment. (B) Increased cell proliferation from sh-Srebf2 is reversed by GW6471. (C, D) GW6471 negates decreased cell apoptosis and altered Bax/Bcl-2 ratios in cells with sh-Srebf2. (D) PCNA upregulation in LPS-treated Caco-2 cells with sh-Srebf2 is reversed by GW6471. (E) Decreased levels of inflammatory cytokines IL-1 β , IL-6, and TNF- α due to sh-Srebf2 are reversed by GW6471. Data are presented as mean $\pm$ standard deviation (SD). $n = 3$. ** $P < 0.01$, *** $P < 0.001$ vs LPS + sh-NC group; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs LPS + sh-Srebf2 group.

4.6. *Srebf2* inhibits cell proliferation, promotes apoptosis, and facilitates the secretion of inflammatory factors by inhibiting the PPAR signaling pathway

To investigate the mitigating effect of *Srebf2* knockdown on inflammation and its relationship with the PPAR signaling pathway, we transfected LPS-treated Caco-2 cells with sh-*Srebf2* and intervened with GW6471, a PPAR signaling pathway inhibitor. As shown in Fig. 6A, PPAR levels were elevated in LPS-treated Caco-2 cells with sh-*Srebf2* ($P < 0.001$), but this elevation was reversed by GW6471 ($P < 0.001$). Cell proliferation increased with *Srebf2* knockdown in LPS-treated Caco-2 cells ($P < 0.001$), but this increase was reversed by GW6471 ($P < 0.001$, Fig. 6B). These findings suggest that *Srebf2* inhibits cell proliferation by suppressing the PPAR signaling pathway. Apoptosis decreased, as indicated by reduced Bax levels and increased Bcl-2 levels in LPS-treated Caco-2 cells with sh-*Srebf2* ($P < 0.001$), though these alterations were reversed by GW6471 ($P < 0.001$, Fig. 6C, D), suggesting that *Srebf2* promotes apoptosis by inhibiting the PPAR signaling pathway. Additionally, PCNA levels were upregulated in LPS-treated Caco-2 cells with sh-*Srebf2* ($P < 0.001$), but this upregulation was reversed by GW6471 ($P < 0.001$, Fig. 6D). The levels of inflammatory markers IL-1 β , IL-6, and TNF- α significantly decreased in LPS-treated Caco-2 cells with sh-*Srebf2* ($P < 0.001$), but this reduction was reversed by GW6471 ($P < 0.001$, Fig. 6E), suggesting that *Srebf2* promotes the secretion of cellular inflammatory factors by inhibiting the PPAR signaling pathway. These results indicate that in LPS-treated Caco-2 cells, *Srebf2* inhibits cell proliferation, promotes apoptosis, and facilitates the secretion of inflammatory factors by inhibiting the PPAR signaling pathway.

5. Discussion

Perianal inflammation is typically caused by various factors, including infection and inflammatory bowel disease, and can significantly impact patients' quality of life [28]. Current treatments for perianal inflammation, such as corticosteroids and immunosuppressants, have several limitations [29]. Therefore, exploring new therapeutic agents that offer strong efficacy with minimal side effects is essential. Sitosterol is a plant-derived sterol that has emerged as a potential anti-inflammatory agent [9]. Our study first identified Sitosterol as the primary active ingredient in the anti-inflammatory compound through a web-based pharmacological screening. We found that *Srebf2* is a key target gene of Sitosterol, primarily clustered in the PPAR signaling pathway. Based on these findings, the present study aimed to explore the role of Sitosterol in alleviating perianal inflammation and its mechanism of action, and its effect on the gut microbiota. By employing in vivo and in vitro models, we aimed to elucidate the effects of Sitosterol on inflammation, cellular mechanisms, and microbiota dynamics, focusing specifically on its interaction with the *Srebf2* and PPAR signaling pathways.

Sitosterol is believed to have anti-inflammatory effects on various of inflammatory diseases, including neuroinflammation, colitis, and allergic asthma [10,11,30]. However, studies on its effects in perianal ulcerative inflammatory diseases have not been reported. Our study found that Sitosterol ameliorated perianal ulcers by reducing LPS-induced inflammatory factor secretion from Caco-2 cells and decreasing serum levels of inflammatory factors in rats with perianal ulcers. Cell proliferation is essential for tissue repair and regeneration. When tissue damage triggers inflammation, surrounding cells proliferate to replenish the damaged cells, thereby restoring the normal function of the tissue [31,32]. The process of apoptosis can be induced by inflammatory mediators and cytokines produced in response to inflammation. Conversely, excessive apoptosis in tissues can trigger further inflammation due to the ineffective removal of apoptotic cells [33]. Our study found that Sitosterol promotes cell proliferation and inhibits apoptosis, suggesting that it may play a role in cell proliferation, tissue repair, and the removal of apoptotic cells, thereby helping to alleviate

inflammation.

Perianal ulcers and abscesses are common symptoms of Perianal Crohn's disease (CD) [34], which is a type of inflammatory bowel disease (IBD) associated with alterations in the gut microbiota [35]. Therefore, it is crucial to examine the role of Sitosterol in mitigating perianal inflammation in relation to gut microbiota composition. An imbalance in gut microbial homeostasis is believed to contribute to both local and systemic inflammation in the body, thereby promoting the development of IBD [36,37]. Our study found that Sitosterol induced the restoration of intestinal microbiota in rats with perianal inflammation. This finding is consistent with the study by Liyuan Ma et al. in 2023 [13], which also indicated that Sitosterol has the effect of restoring intestinal microbiota. We conclude that Sitosterol alleviates perianal inflammation in rats by restoring the composition of the intestinal flora.

In exploring the molecular mechanism of Sitosterol in the treatment of perianal inflammation, our study first identified *Srebf2* as the target gene of Sitosterol through network pharmacology. We found that *Srebf2* is clustered within the PPAR signaling pathway. While *Srebf2* has been reported to activate NLRP3 inflammatory vesicles, leading to endothelial inflammation and atherosclerosis formation [16], our study revealed that Sitosterol inhibited the expression of *Srebf2* in rats with perianal ulcers. To further investigate the relationship between *Srebf2* expression inhibition and perianal inflammation, we knocked down *Srebf2* expression in both perianal ulcerated rats and LPS-induced Caco-2 cells. The results indicate that knocking down *Srebf2* restores gut flora and alleviates inflammation in rats with perianal ulcers. This finding aligns with a study by Wen Chen et al. in 2022, which demonstrated that an imbalance in gut flora upregulates *Srebf2* [25]. In the LPS-induced cellular inflammation model, the knockdown of *Srebf2* led to significant cell proliferation, inhibition of apoptosis, and reduced secretion of cellular inflammatory factors. Additionally, PPAR, a nuclear receptor responsible for cellular energy metabolism [38], interferes with the activation and function of NF- κ B during the inflammatory response, thereby inhibiting inflammation [39]. To further investigate the relationship between *Srebf2* and the PPAR signaling pathway, the study introduced the PPAR pathway inhibitor GW6471 to validate the role of *Srebf2* knockdown in alleviating inflammation. The results showed that GW6471 could enhance the secretion of cellular inflammatory factors by inhibiting cell proliferation and promoting apoptosis. This finding suggests that the inhibition of *Srebf2* expression may activate the PPAR pathway, thereby improving perianal inflammation.

In conclusion, the present study highlights Sitosterol as a potent anti-inflammatory agent that restores the composition of intestinal microorganisms and alleviates perianal inflammation in rats with perianal inflammatory conditions. This effect is achieved by inhibiting the expression of *Srebf2* and activating the PPAR pathway. These findings provide a solid foundation for further exploration of Sitosterol as a therapeutic agent and offer insights into the interactions between plant-derived compounds, inflammation, and gut health. Future studies could investigate specific interactions within the PPAR pathway and explore the potential clinical applications of Sitosterol in inflammatory conditions.

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Ethical approval

The ethics committee of Animal Ethics Committee of Hangzhou

Hebei Technology Co., Ltd. approved the animal study (ZJLL-20230106).

CRediT authorship contribution statement

Yanlan Wu: Writing – original draft, Data curation. **Hao Ge:** Investigation, Formal analysis. **Haoran Zhao:** Methodology, Investigation. **Kaiping Zou:** Methodology, Investigation. **Pei Wang:** Methodology, Investigation. **Yi Wang:** Writing – review & editing. **Yang Zhang:** Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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